**NXP Semiconductors**
User's Guide

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# Getting Started with MCUXpresso SDK CMSIS Packs

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## 1 Introduction

The MCUXpresso Software Development Kit (SDK) is a comprehensive software enablement package designed to simplify and accelerate application development with NXP's microcontrollers based on Arm® Cortex®-M cores. The MCUXpresso SDK includes production-grade software with integrated RTOS (optional), integrated stacks and middleware, reference software, and more.

For more information about MCUXpresso SDK please visit MCUXpresso SDK homepage.

MCUXpresso SDK is provided to users in two ways:

- SDK packages downloaded from MCUXpresso SDK builder homepage
- CMSIS packs downloaded from MCUXpresso packs repository

This document describes how to download MCUXpresso SDK CMSIS packs, how to open demo application and create new project using the packs.

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# 2 MCUXpresso SDK CMSIS packs introduction

## 2.1 CMSIS pack standard

The CMSIS pack represents delivery mechanism for software components, devices and evaluation boards support. CMSIS pack standard is defined by Arm and supported by various IDE vendors including Keil uVision and IAR Embedded Workbench for Arm.

CMSIS standard recognizes three main types of CMSIS packs:

- Pack with software components (Software Pack, SWP)
- Pack with device support (Device Family Pack, DFP)
- Pack with board support (Board Support Pack, BSP)

Software components (CMSIS components) are a collection of source modules, header and configuration files and libraries. They can be used in all three types of CMSIS packs. Each CMSIS component has unique identification that serves also for components organization in IDE CMSIS component managers: CMSIS component vendor, bundle, class, group, subgroup, variant, and version.

Content of CMSIS packs including CMSIS components is described by XML-based package description (PDSC) file. The PDSC file is designed for software development environments and describes the user and device relevant context for the files supplied within a pack.

For more information about CMSIS pack standard please visit CMSIS Pack Documentation homepage.

## 2.2 CMSIS packs with MCUXpresso SDK overview

The MCUXpresso SDK is delivered in Device Family Packs and Board Support Packs.

### 2.2.1 MCUXpresso SDK Device Family Packs

The Device Family Pack provides minimal device support. Installing DFP provides enough resources to create, compile and debug MCUXpresso SDK project.

MCUXpresso SDK Device Family Pack contains:

- Device header files and system initialization modules
- Startup files
- Linker files
- SVD files
- Flash drivers (for some of the development tools)
- SDK drivers and utilities
- SDK project templates

Content of the Device Family Pack is organized in following CMSIS components:

- Startup, contains minimal set of sources needed to create and compile MCUXpresso SDK project
- SDK drivers

- SDK utilities
- SDK CMSIS drivers
- RTE Device SDK project template, contains RTE_Device.h file required by CMSIS drivers
- Board SDK project template, contains board, clock configuration, peripherals and pin mux SDK configuration files with general implementation

### 2.2.2 MCUXpresso SDK Board Support Packs

The Board Support Pack provides support for NXP evaluation board and example projects for the board.

MCUXpresso SDK Board Support Pack contains:

- Example projects and demo applications
- SDK project templates containing board, clock configuration, peripherals and pin mux SDK configuration files implemented for the specific board

# 3 Install MCUXpresso SDK CMSIS pack

This section describes installation of MCUXpresso SDK CMSIS packs in Keil® MDK/µVision and IAR Embedded Workbench for Arm. MK64F12 Device Family Pack is used as an example, however, following steps can be used to install any MCUXpresso SDK CMSIS packs.

**NOTE**

The MCUXpresso SDK CMSIS packs uses ARM::CMSIS pack. Make sure that the ARM::CMSIS pack is installed in your development environment. In Keil MDK/µVision, this pack is installed by default. To install it in IAR Embedded Workbench for Arm, go to *Project* > *CMSIS-Packs* > *Pack Installer*. In the *CMSIS Pack Manager* window, go to *ARM* > *CMSIS* and install the Arm CMSIS pack. See the *MCUXpresso SDK CMSIS Packs Release Notes* for the version of Arm CMSIS pack to use.

## 3.1 Install MCUXpresso SDK CMSIS pack using Keil® MDK/µVision

1. Open *Pack Installer*. Go to *Project* > *Manage* > *Pack Installer* or use the toolbar:

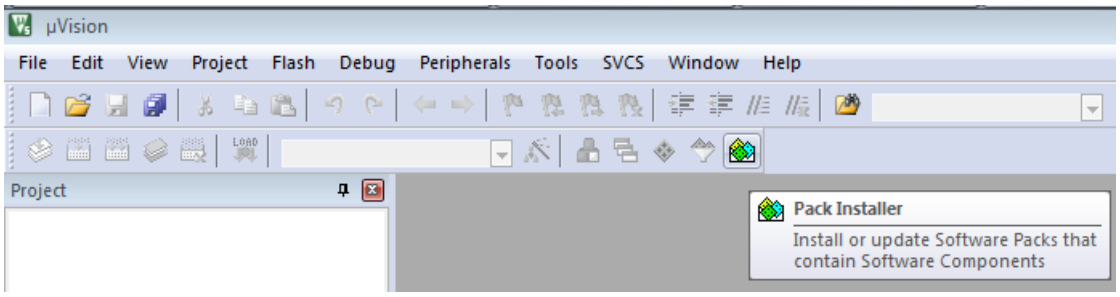



**Figure 1. Running Pack Installer from Keil MDK/µVision toolbar**

2. In the Pack Installer, on the left pane you can switch between *Devices* and *Boards* view. In the *Devices* view, select *NXP* > *MK64F12* or type MK64F12 into *Search* field on top.

3. On the right pane, you can switch between *Packs* and *Examples* view. In the *Packs* view, you will see CMSIS packs related to device or board selected on the left pane. Find *NXP::MK64F12_DFP* and click on the *Install* button next to it.

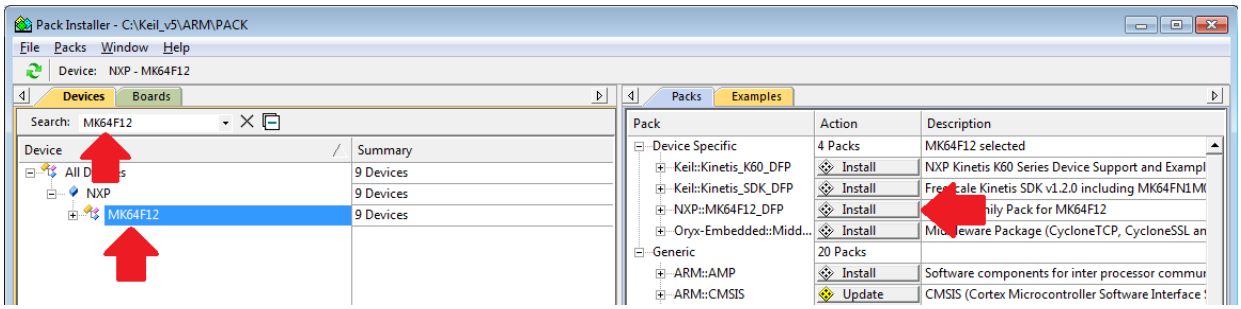



**Figure 2. Pack installation in Keil MDK/µVision Pack Installer**

4. Before pack installation, you must accept the license agreement. After accepting the license terms, the pack is installed.

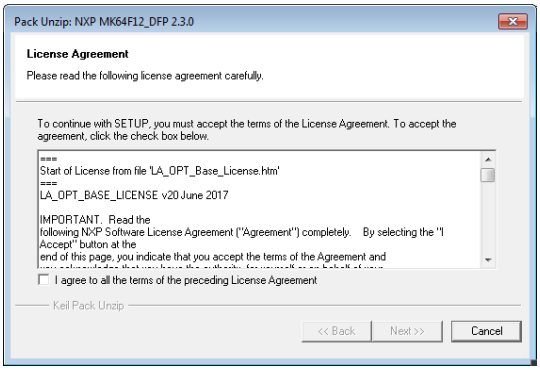



**Figure 3. License agreement dialog**

5. Close Pack Installer.

## 3.2 Install MCUXpresso SDK CMSIS pack using IAR Embedded Workbench for Arm

1. Add Keil CMSIS pack repository to *Remote Pack Repositories*. Go to *Tools > Options* and in the CMSIS Pack options add http://www.keil.com/pack to *Remote Pack Repositories*. Click *OK*.

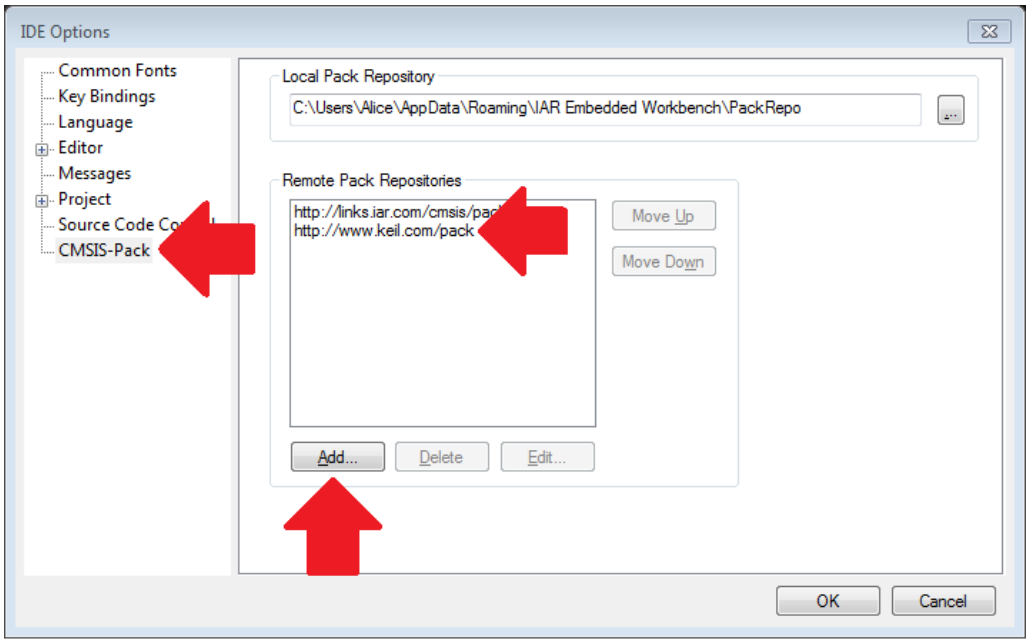



**Figure 4. Remote Pack Repositories in IAR Embedded Workbench for Arm tool options**

2. Open *Pack Installer*. Go to *Project* > *CMSIS-Packs* > *Pack Installer* or use the toolbar:

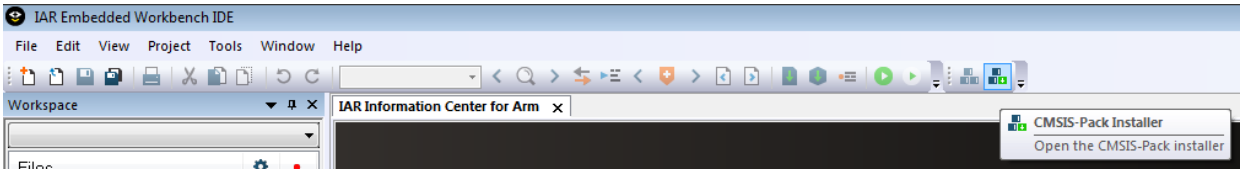



**Figure 5. Running Pack Installer from IAR Embedded Workbench for Arm toolbar**

3. In the *Pack* view, select *NXP* > *MK64F12_DFP*.
4. Expand the *MK64F12_DFP* to see available versions of this pack. Right click the version 2.3.0 and click *Install* in the pop-up menu.

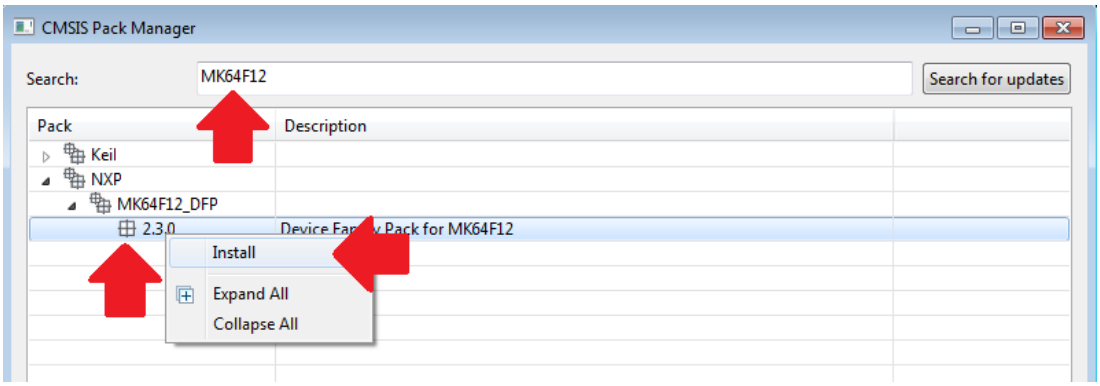



**Figure 6. Pack installation in IAR Embedded Workbench for Arm Pack Installer**

5. Close the Pack Installer.

# 4 Open a demo application

This section describes how to open a demo application from MCUXpresso SDK Board Support Pack in Keil MDK/µVision and IAR Embedded Workbench for Arm. Hello World demo from FRDM-K64F Board Support Pack is used to demonstrate the process. Same steps can be used to open all example projects in MCUXpresso SDK Board Support Packs.

## 4.1 Open a demo application using Keil MDK/µVision

1. Go to *Project* > *Manage* > *Pack Installer* and install NXP::MK64F12_DFP and NXP::FRDM-K64F_BSP packs. Make sure that ARM::CMSIS pack is also installed.
2. In the *Pack Installer*, on the left pane select the *Boards* view and select the FRDM-K64F board. You can use the *Search* field at the top of the window to filter the board. To make sure the correct BSP is selected, hover mouse over the board name – NXP.FRDM-K64F_BSP pack should be shown as the parent pack.

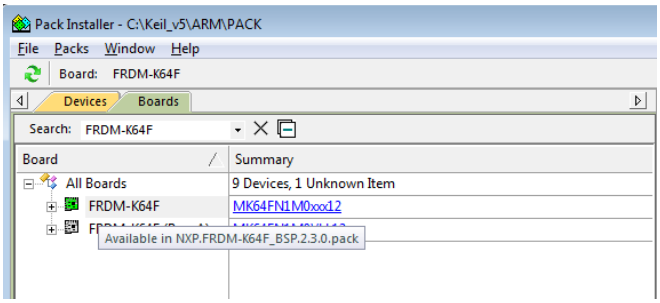



**Figure 7. Board selection in Keil MDK/µVision Pack Installer**

3. On the right pane, select the *Examples* view and find the hello_world example. Click the *Copy* button next to it.

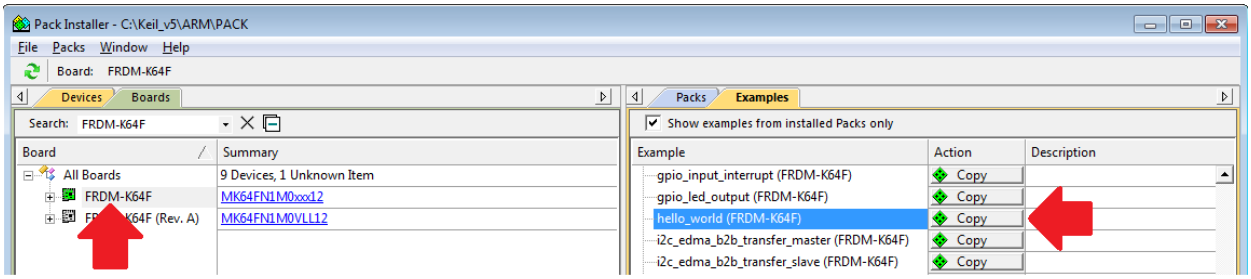



**Figure 8. Selecting example from pack in Keil MDK/µVision Pack Installer**

4. Choose the directory to copy the demo application. If you want to keep the directory structure from the pack, select the *Use Pack Folder Structure* checkbox. Select the *Launch µVision* checkbox to automatically open the copied project. Click the *OK* button.

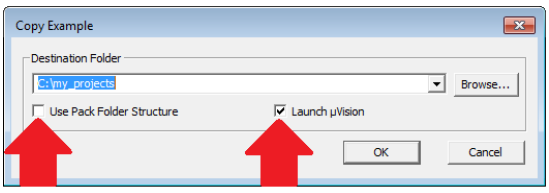


**Figure 9. Copying example from pack in Keil MDK/µVision**

5. A new instance of the µVision with the Hello World demo application is opened. Now you can build and run the demo.

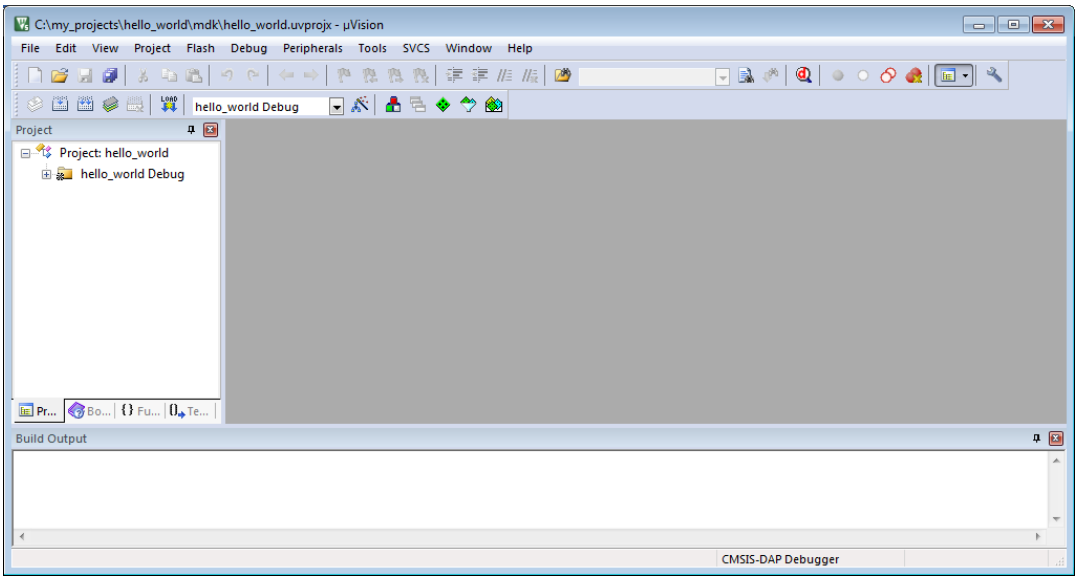



**Figure 10. Example project from pack in Keil MDK/µVision**

## 4.2 Open a demo application using IAR Embedded Workbench for Arm

1. Go to *Project* > *CMSIS-Packs* > *Pack Installer* and install NXP MK64F12_DFP and FRDM-K64F_BSP packs. Make sure that Arm CMSIS pack is also installed.
2. Go to *Project* > *Create New Project…*, select *CMSIS Pack example* and click the *OK* button.

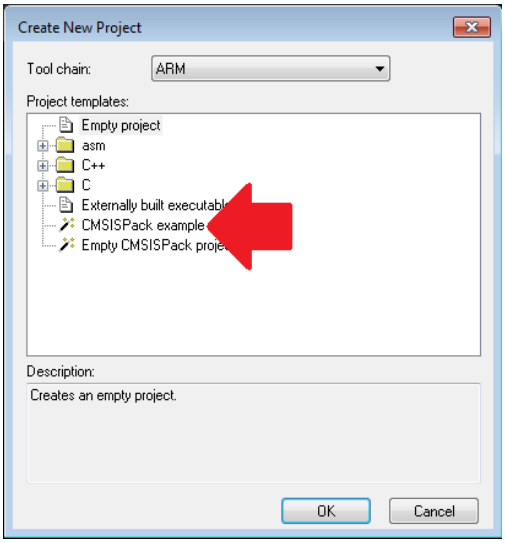



**Figure 11. Creating new project from example in pack in IAR Embedded Workbench for Arm**

3. Go to *NXP* > *MK64F12* > *MK64FN1M0xxx12* and select one of the package variants. You can also use the *Search* filed above the device selector. Click the *Next* button.

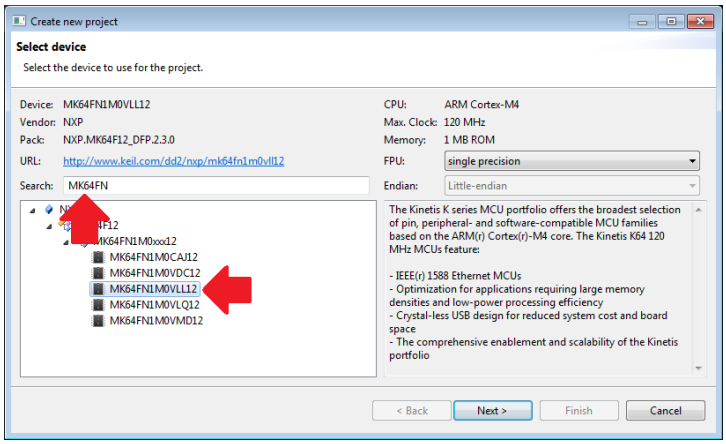



**Figure 12. Creating new project from example in pack in IAR Embedded Workbench for Arm**

4. Select the *hello_world* example from *demo_apps/hello_world*. Click the *Finish* button.

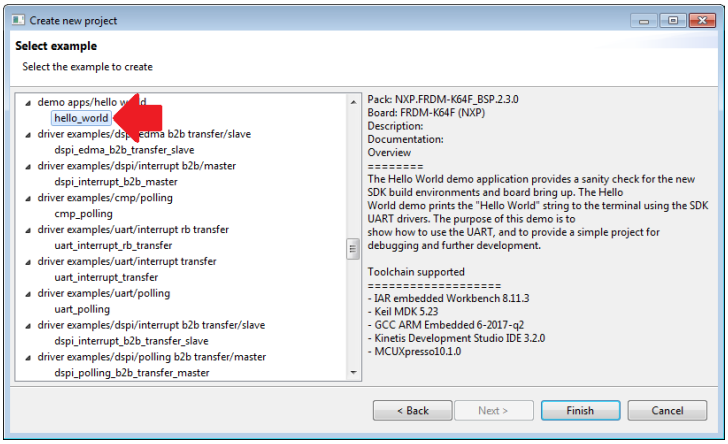



**Figure 13. Selecting example from pack in IAR Embedded Workbench for Arm**

5. Choose the directory to copy the demo application and click the *OK* button.
6. The Hello World demo application is opened. Now you can build and run the demo.

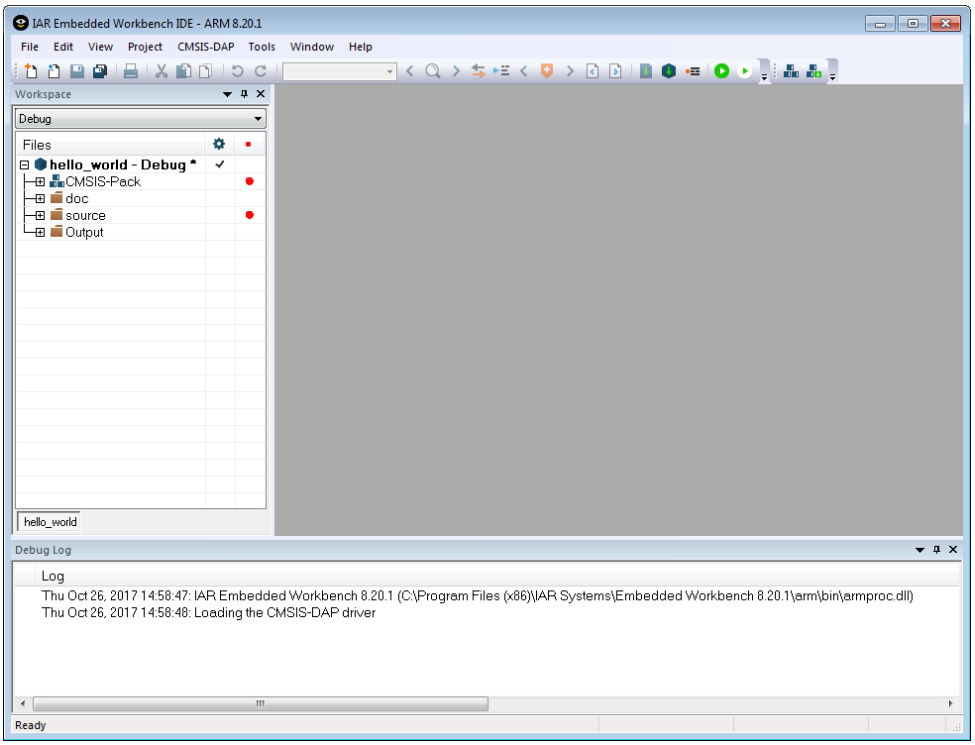



**Figure 14. Example project from pack in IAR Embedded Workbench for Arm**

# 5 Create a new application

This section describes how to create a new project using MCUXpresso SDK CMSIS packs in Keil MDK/µVision and IAR Embedded Workbench for Arm. New project for MK64F1M0VLL12 device from MK64F12_DFP pack is used to demonstrate the process, however, same steps can be used to create new project for all devices supported in MCUXpresso SDK Device Family Packs.

## 5.1 Create a new application using Keil MDK/µVision

1. Go to *Project* > *Manage* > *Pack Installer* and install NXP::MK64F12_DFP. Make sure that ARM::CMSIS pack is also installed.
2. Go to *Project* > *New µVision Project…* and select folder to create the new project.
3. Go to *NXP* > *MK64F12* and select one of the device package variant, for example *MK64F1M0VLL12*. You can also use *Search* field to find devices.

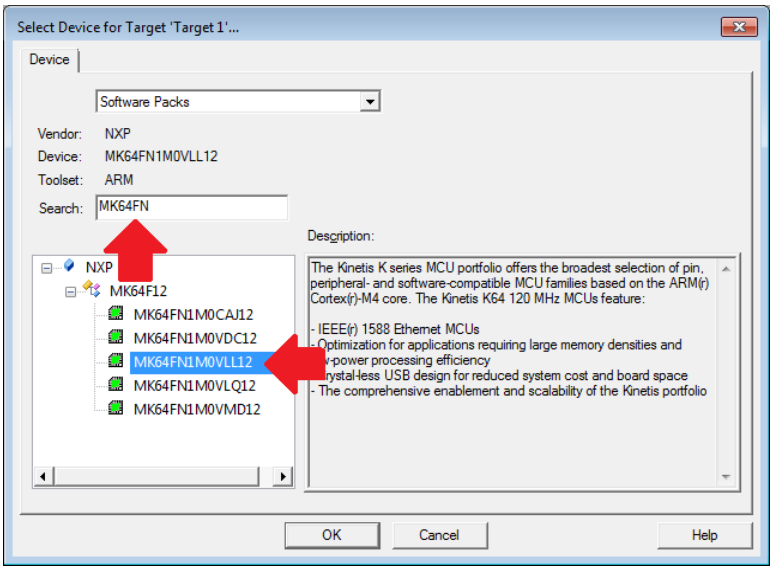



**Figure 15. Device selection in Keil MDK/µVision**

4. Open *Manager Run-Time Environment* in *Project > Manage > Run-Time Environment*. Here you can add and remove components in the project.

    a. To create basic MCUXpresso SDK project, enable *Device > Startup* component. This component contains basic set of files for a new project. To use it, you also have to enable *CMSIS > CORE* component, which is highlighted in yellow color as dependency for the *Device > Startup*. Enable the *CMSIS > CORE* component manually or click the *Resolve* button to resolve the dependency automatically.

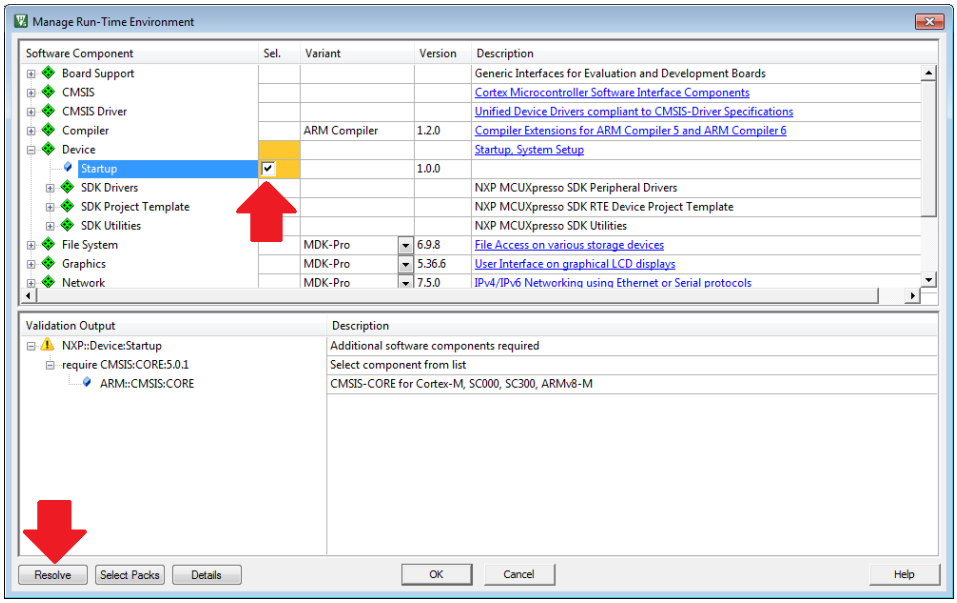



**Figure 16. Creating basic MCUXpresso SDK project in Keil MDK/µVision**

    b. To create a new project with minimal set of MCUXpresso SDK components, enable *Board Support > SDK Project Template > project_template* component. This component

contains board configuration files used by MCUXpresso SDK and will require additional MCUXpresso SDK components needed to create a simple Hello World application.

After project_template component is enabled, number of missing components is reported. Click *Resolve* button to add missing MCUXpresso SDK components required to build simple MCUXpresso SDK project.

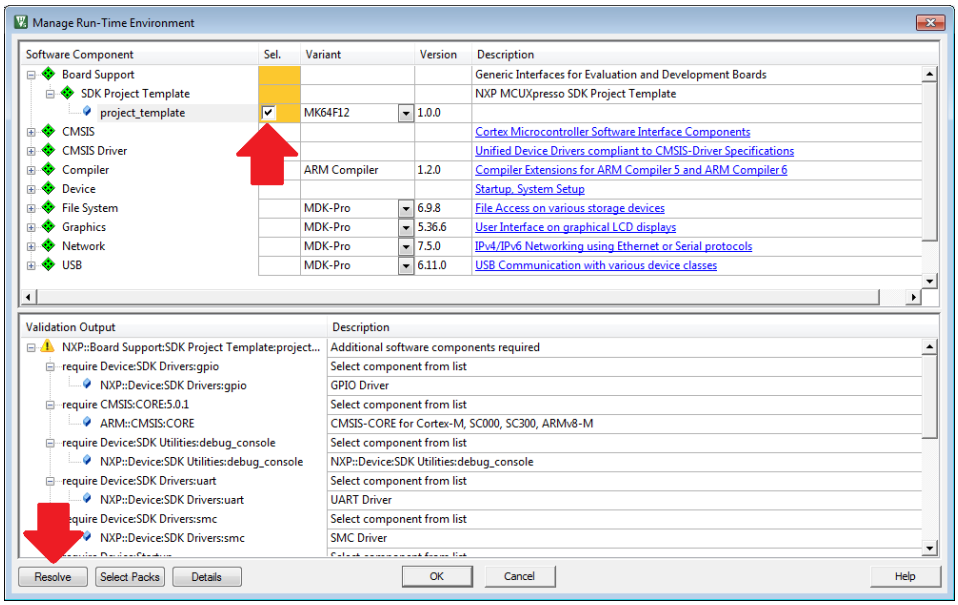



**Figure 17. Creating MCUXpresso SDK project in Keil MDK/µVision**

c. If you install also FRDM-K64F_BSP, you can select project template with board configuration files preconfigured for that specific board by switching variant of the project template component.

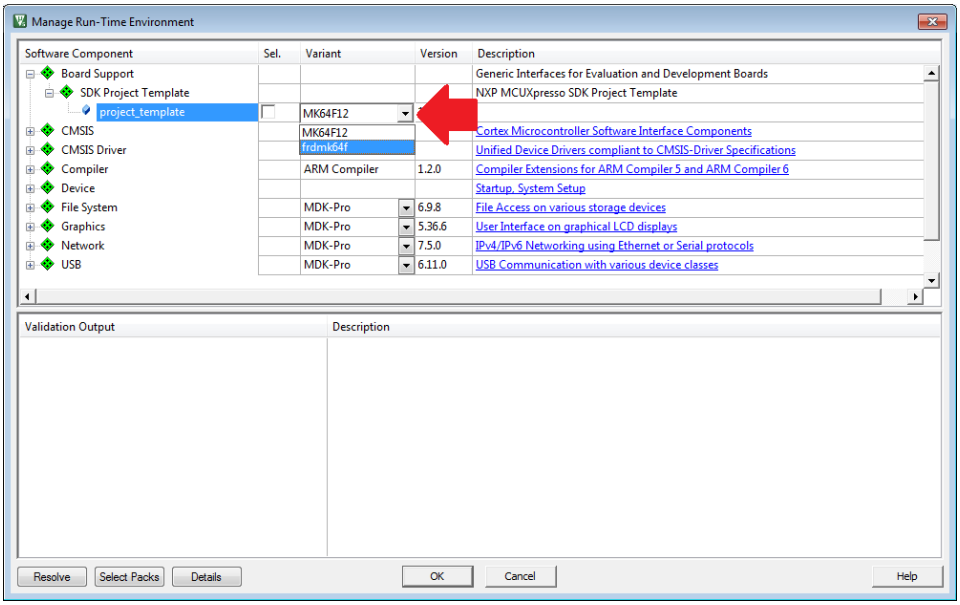



**Figure 18. Selecting board specific MCUXpresso SDK project template in Keil MDK/µVision**

5. Click *OK* to close *Run-Time Environment Manager*.
6. Add the main function. Right click on the *Source Group 1* and select *Add New Item to Group*. Select *C File (.c)* and add "main" to *Name*. Click *Add*. A new file *main.c* was added to the project where you can add *main()* function of your application.

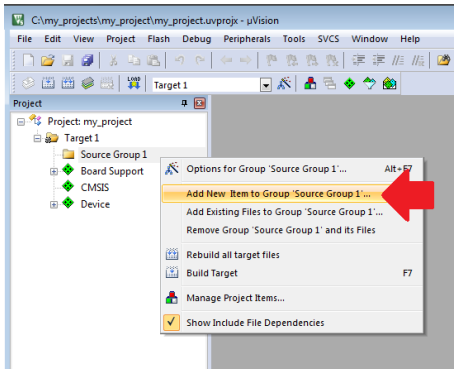



**Figure 19. Adding new file in Keil MDK/µVision**

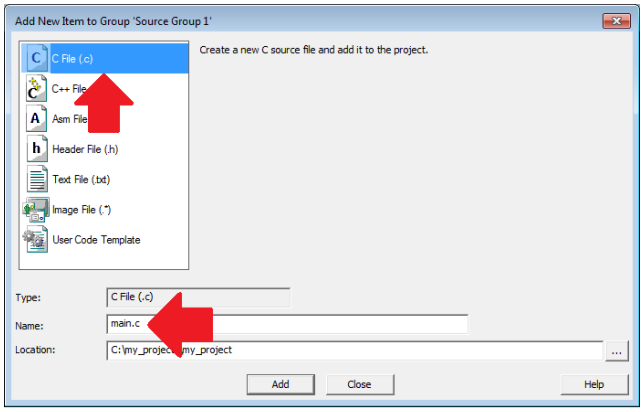



**Figure 20. Creating main.c file in Keil MDK/µVision**

7. To properly link MCUXpresso SDK project, scatter file included in the MCUXpresso SDK pack must be used. To make sure that the scatter file is not generated by Keil MDK/µVision and file from pack is used instead, go to *Project > Options for Target > Linker* and make sure that the option *Use Memory Layout from Target Dialog* is not enabled.

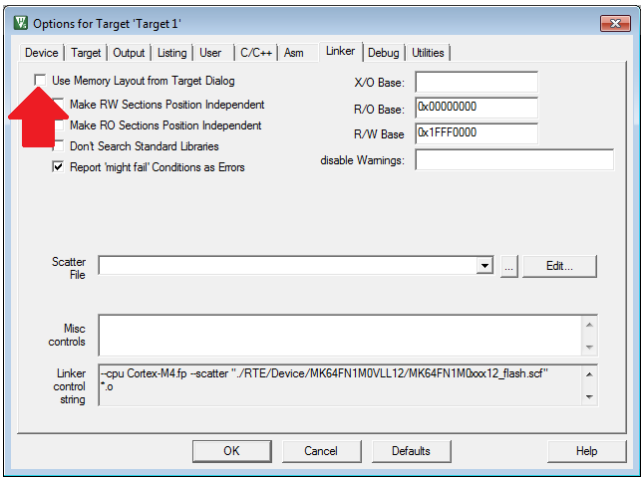



**Figure 21. Disabling generated scatter file in Keil MDK/µVision**

8. You can build and debug your application now.

## 5.2 Create a new application using IAR Embedded Workbench for Arm

1. Go to *Project* > *CMSIS-Packs* > *Pack Installer* and install NXP MK64F12_DFP. Make sure that Arm CMSIS pack is also installed.
2. Go to *Project* > *Create New Project…*, select *Empty CMSIS Pack project* and click the *OK* button.

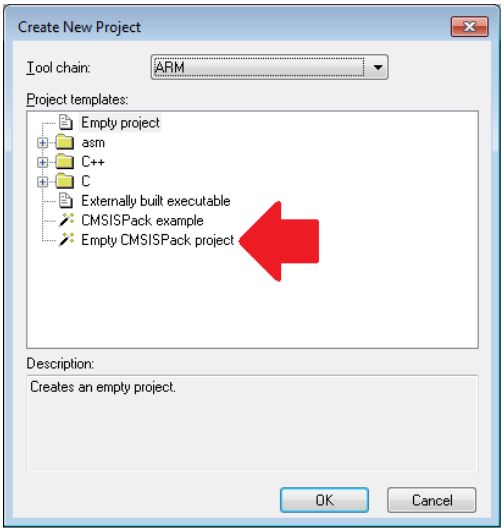



**Figure 22. Creating new project in IAR Embedded Workbench for Arm**

3. Select *NXP* > *MK64F12* > *MK64FN1M0xxx12* > *MK64F1M0VLL12*. You can also use the *Search* filed above the device selector. Click the *Next* button.

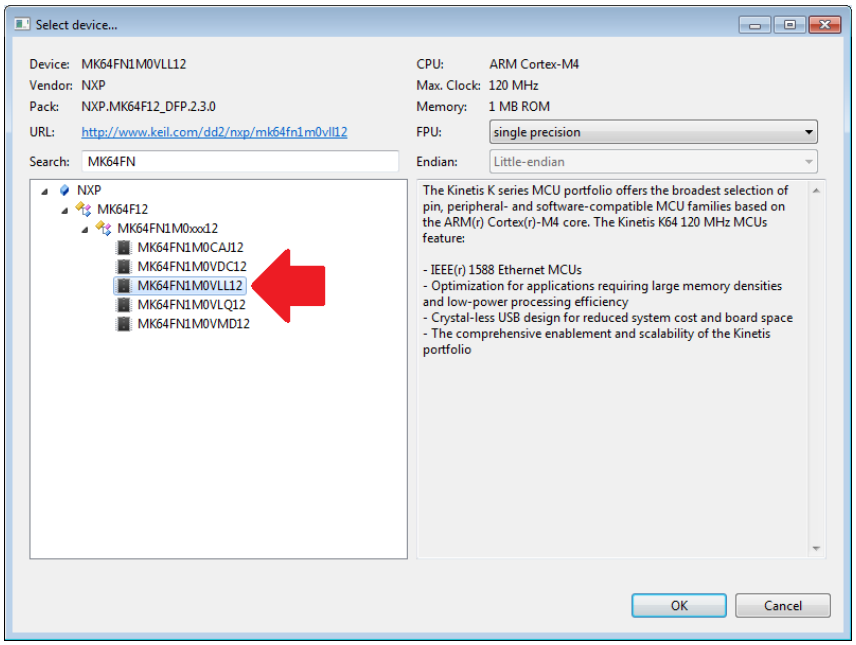



**Figure 23. Device selection in in IAR Embedded Workbench for Arm**

4. Open *Project CMSIS Component Manager* in *Project* > *CMSIS-Pack* > *Project Components*. Here you can add and remove components in the project.

    a. To create basic MCUXpresso SDK project, enable *Device* > *Startup* component. This component contains basic set of files for a new project. To use it, you also have to enable *CMSIS* > *CORE* component, which is highlighted in yellow color as dependency for the *Device* > *Startup*. Enable the *CMSIS* > *CORE* component manually or click the *Resolve* button to resolve the dependency automatically.

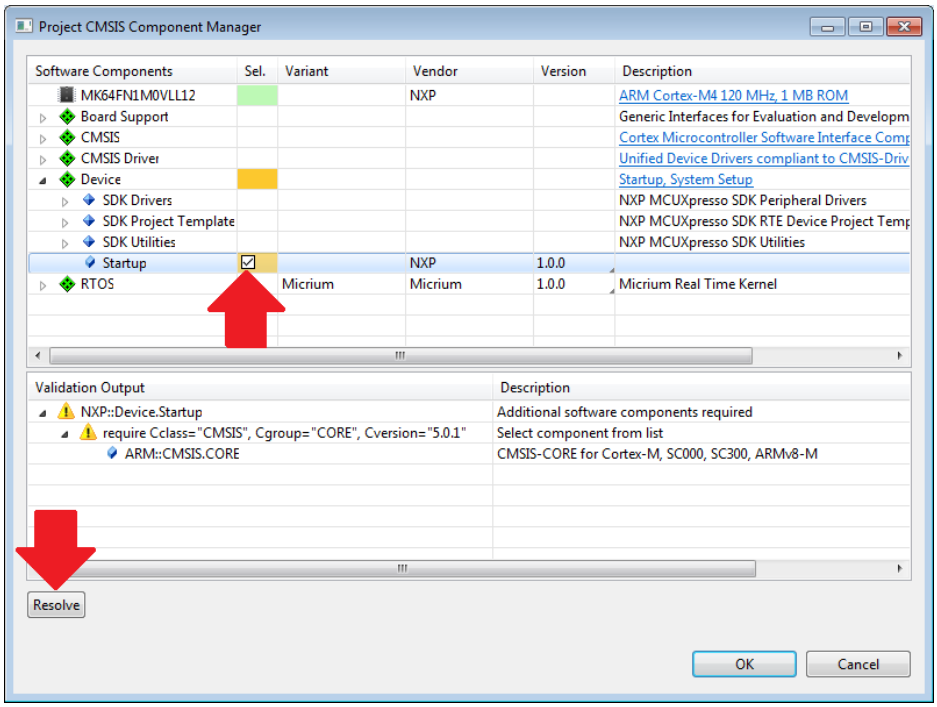


**Figure 24. Creating basic MCUXpresso SDK project in IAR Embedded Workbench for Arm**

b. To create a new project with minimal set of MCUXpresso SDK components, enable *Board Support > SDK Project Template > project_template* component. This component contains board configuration files used by MCUXpresso SDK and will require additional MCUXpresso SDK components needed to create a simple Hello World application.

After project_template component is enabled, number of missing components is reported. Click the *Resolve* button to add missing MCUXpresso SDK components required to build simple MCUXpresso SDK project.

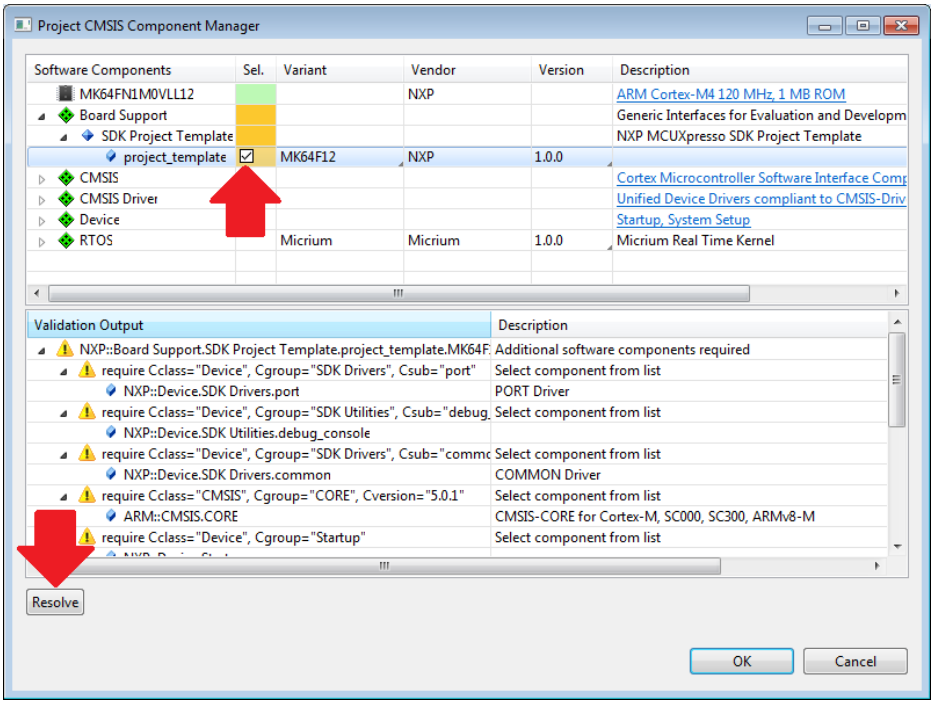



**Figure 25. Creating MCUXpresso SDK project in IAR Embedded Workbench for Arm**

c. If you also install FRDM-K64F_BSP, you can select project template with board configuration files preconfigured for that specific board by switching variant of the project template component.

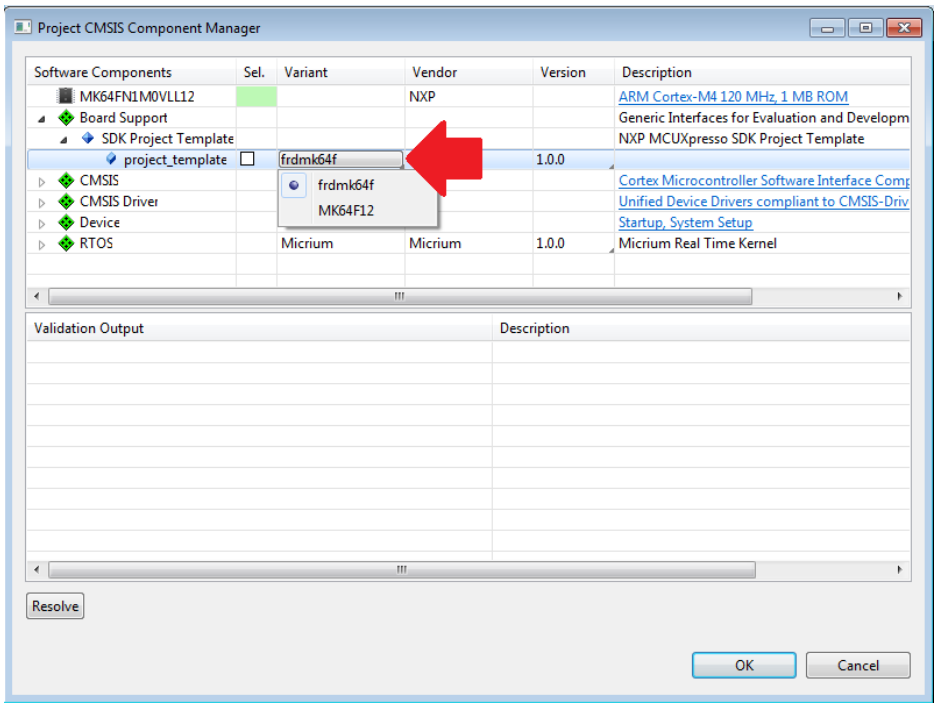



**Figure 26. Selecting board specific MCUXpresso SDK project template in IAR Embedded Workbench for Arm**

5. Click the *OK* button to close *Project CMSIS Component Manager*.
6. Add the main function. Go to *File > New File* and create new *main.c* file where you can add *main()* function of your application. Add the created file to project using *Project > Add Files…*.
7. You can build and debug your application now.

**NOTE**

To debug projects using CMSIS DAP debugger, default flash loader .board file used by IAR Embedded Workbench has to be overridden. In project options, go to *Debugger > Download*, enable *Override default .board file* and select .board file from *{IAR Embedded Workbench installation directory}\arm\config\flashloader\NXP*. Additionally, in *Debugger > CMSIS DAP > Reset* set the reset source to *Software*.

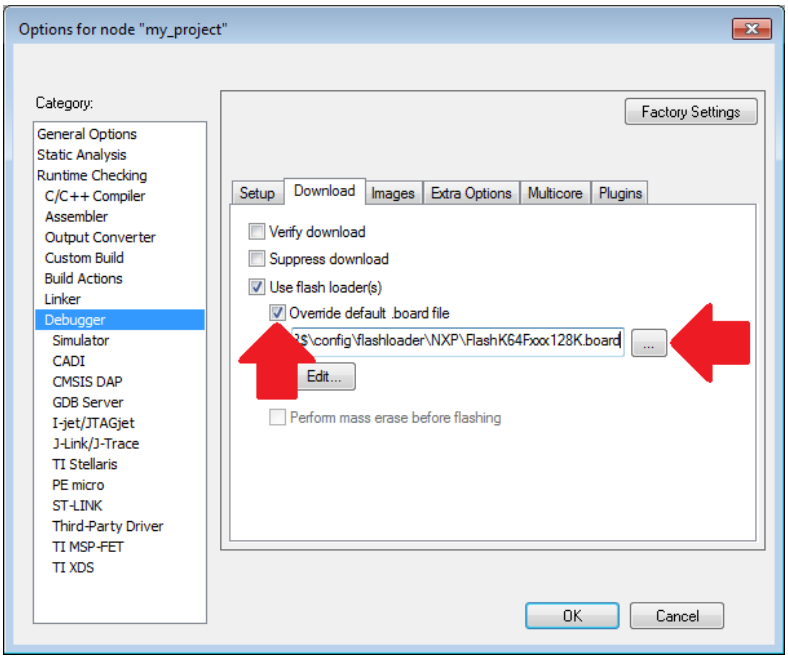



**Figure 27. Overriding default flash loader in IAR Embedded Workbench for Arm**

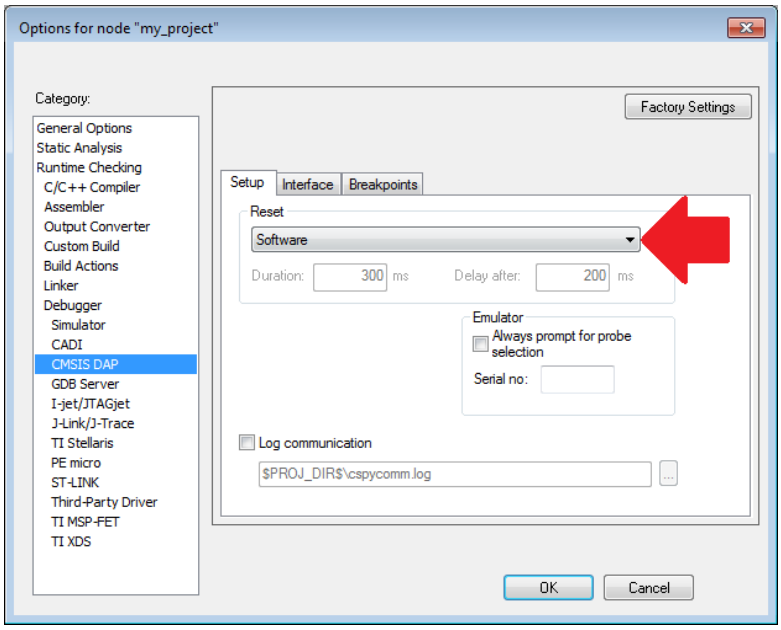



**Figure 28. Setting CMSIS DAP software reset in IAR Embedded Workbench for Arm**

# 6 Convert project created with Keil CMSIS packs to NXP MCUXpresso SDK CMSIS packs

This section describes how to convert an existing project build on CMSIS packs for NXP devices created by Keil. The process will be described on the conversion of a project created for the MK64F1M0xxx12 device from the Keil pack to project using the MK64FN1M0VLL12 device from the MCUXpresso SDK pack in Keil MDK/µVision and IAR Embedded Workbench for Arm. The same

steps can be used to convert projects for all devices supported in MCUXpresso SDK Device Family Packs.

**NOTE**

It is recommended to create a backup copy of the project before conversion.

**NOTE**

Using Keil MDK/µVision, after switching project to device from different CMSIS pack the debugger selected in project debug settings is not preserved. The debugger is automatically set to ULINK and has to be manually restored.

## 6.1 Convert project created with Keil CMSIS packs to NXP MCUXpresso SDK CMSIS packs using Keil MDK/µVision

1. Go to *Project* > *CMSIS-Packs* > *Pack Installer* and install MK64F12_DFP
2. Go to *Project* > *Options for Target* > *Device*
3. You can see *NXP* > *K60 Series* > *MK64FN1M0xxx12* selected in the device selector. Change the selection to *NXP* > *MK64F12* > *MK64FN1M0VLL12*.

   Projects that use CMSIS packs produced by Keil use "Keil." prefix in pack name visible in *Software Pack* > *Pack* and www.keil.com/pack in *Software Pack* > *URL*.

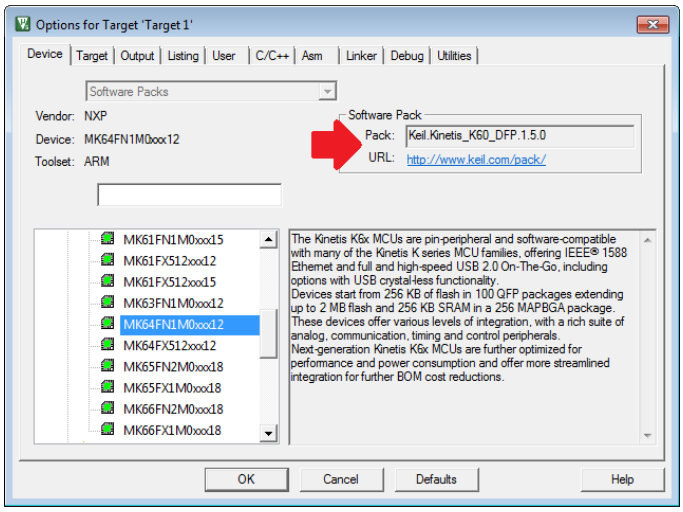



**Figure 29. Device from Keil pack in Keil MDK/µVision**

Projects that use MCUXpresso SDK CMSIS packs produced by NXP use "NXP." prefix in pack name visible in *Software Pack* > *Pack* and mcuxpresso.nxp.com/cmsis_pack/repo in *Software Pack* > *URL*.

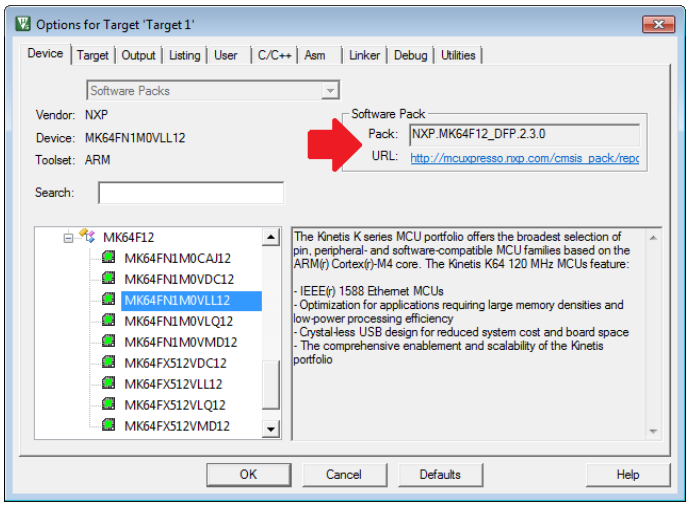



**Figure 30. Device from MCUXpresso SDK pack in Keil® MDK/µVision**

4. In *Options for Target*, switch from *Device* to *Debug* tab. Restore the debugger used in the project.

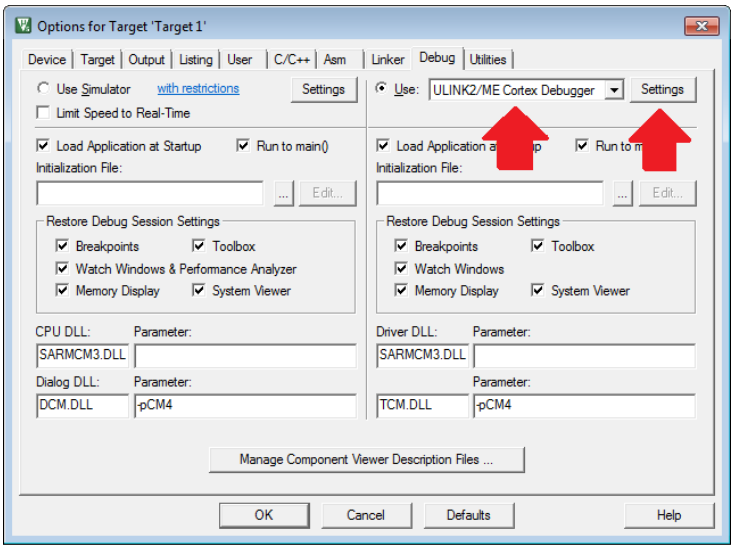



**Figure 31. Debugger settings in Keil MDK/µVision**

5. Click the *OK* button to close *Options for Target*.
6. Open *Run-Time Environment* manager. Here you can see that the *Device* > *Startup* component has switched to component from NXP MK64F12 Device Family Pack. If the component is not switched automatically switch it manually. Distinguish the components by hovering the mouse over the component name. The parent pack is shown in the hint.

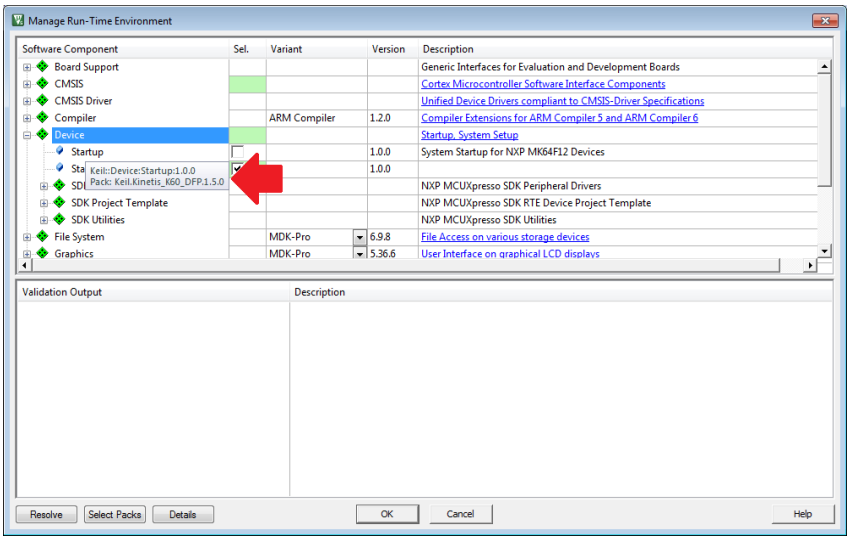



**Figure 32. Component from Keil pack in Keil MDK/µVision**

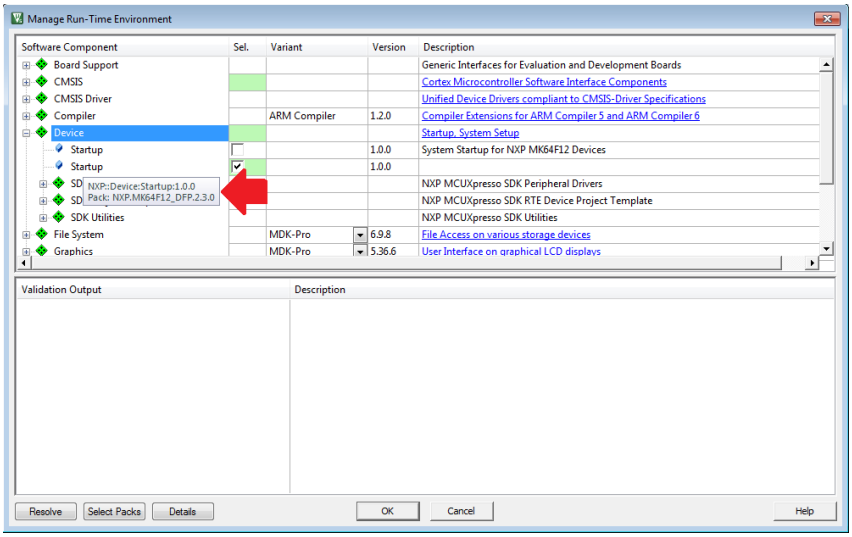



**Figure 33. Component from MCUXpresso SDK pack in Keil MDK/µVision**

7. In the Project Window you can see that Startup component files from NXP MCUXpresso pack replaced the files from Keil pack. Note that the files from Keil pack are still available in the *{project}\RTE\Device\MK64FN1M0xxx12* directory.

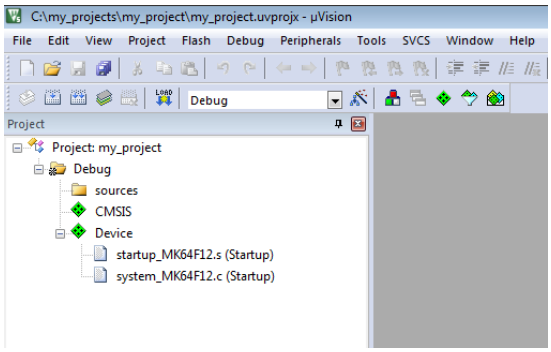



**Figure 34. Startup component from Keil pack in Keil MDK/µVision**

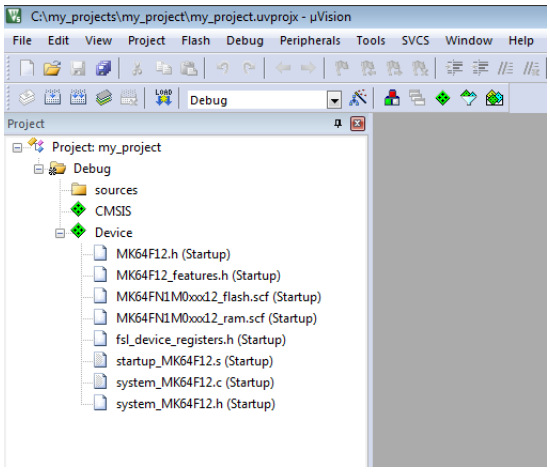



**Figure 35. Startup component from MCUXpresso SDK pack in Keil MDK/µVision**

8. Add more MCUXpresso SDK components using *Run-Time Environment* manager. To make sure MCUXpresso SDK components work properly, it is recommended to use the scatter file included in the MCUXpresso SDK Device Family Pack. To make sure that the scatter file is not generated by Keil MDK/µVision and the file from pack is used instead, go to *Project* > *Options for Target* > *Linker* and make sure that the option *Use Memory Layout from Target Dialog* is not enabled.

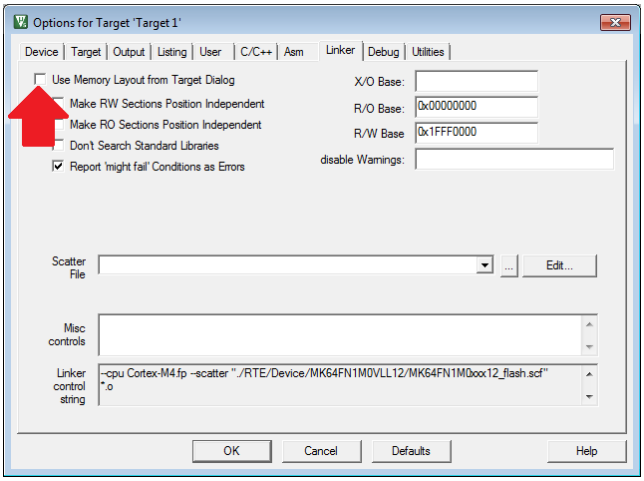



**Figure 36. Disabling generated scatter file in Keil MDK/µVision**

## 6.2 Convert project created with Keil CMSIS packs to NXP MCUXpresso SDK CMSIS packs using IAR Embedded Workbench for Arm

1. Go to *Project* > *Options* > *General Options*. On *Target* tab. Select CMSIS-Pack processor variant and click on button next to it to open device selector.

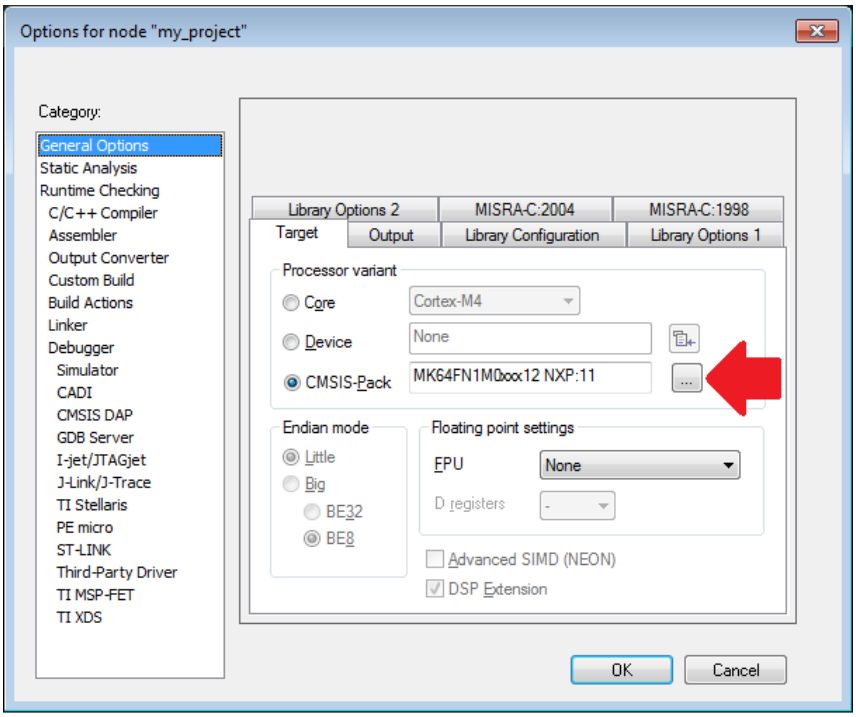



**Figure 37. Changing device in IAR Embedded Workbench for Arm**

2. You can see *NXP > K60 Series > MK64FN1M0xxx12* selected in the device selector. Change the selection to *NXP > MK64F12 > MK64FN1M0xxx12 > MK64FN1M0VLL12*. Click the *OK* button to close device selector.

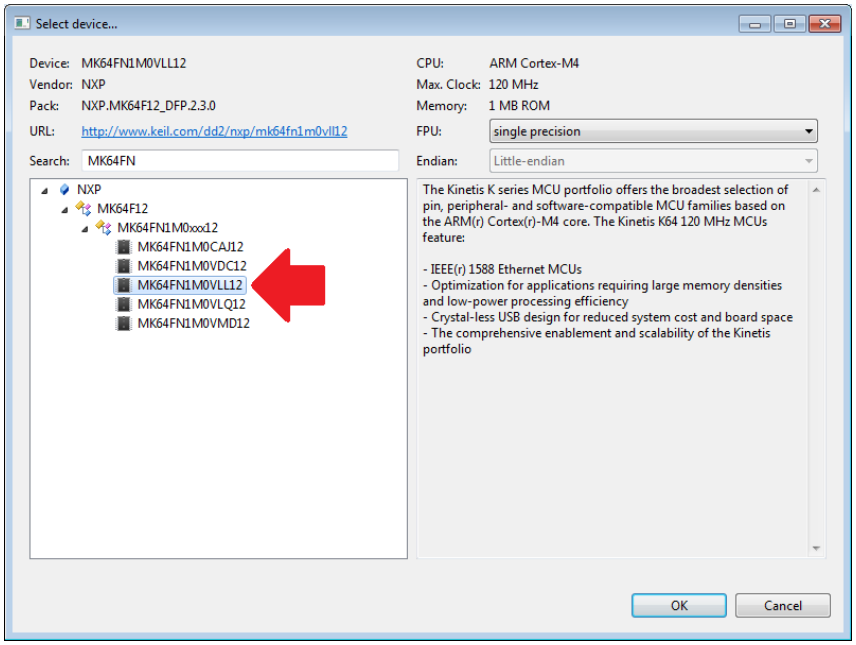



**Figure 38. Device selection in IAR Embedded Workbench for Arm**

Projects that use CMSIS packs produced by Keil use the “Keil.” prefix in pack name visible in the device selector.

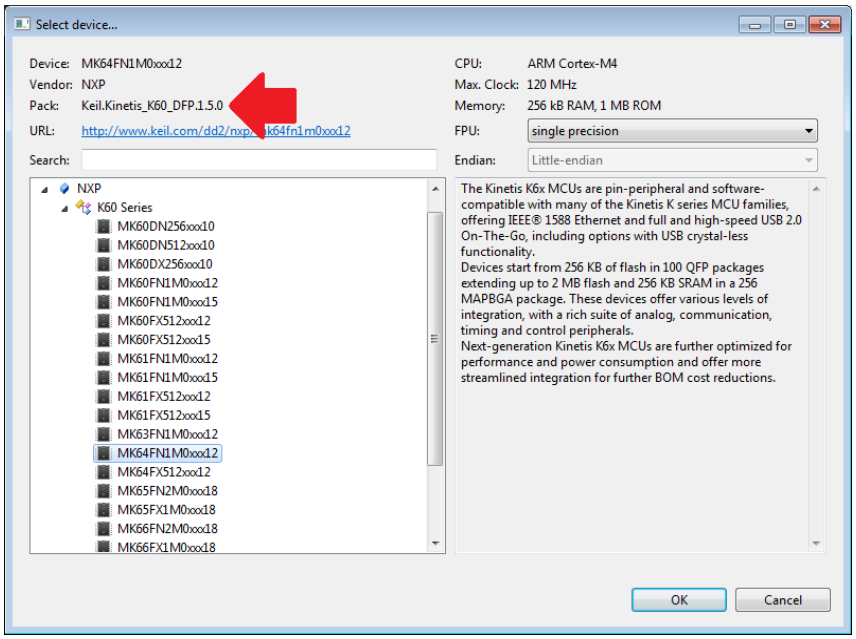



**Figure 39. Device from Keil pack in IAR Embedded Workbench for Arm**

Projects that use MCUXpresso SDK CMSIS packs produced by NXP use the “NXP.” prefix in pack name visible in the device selector.

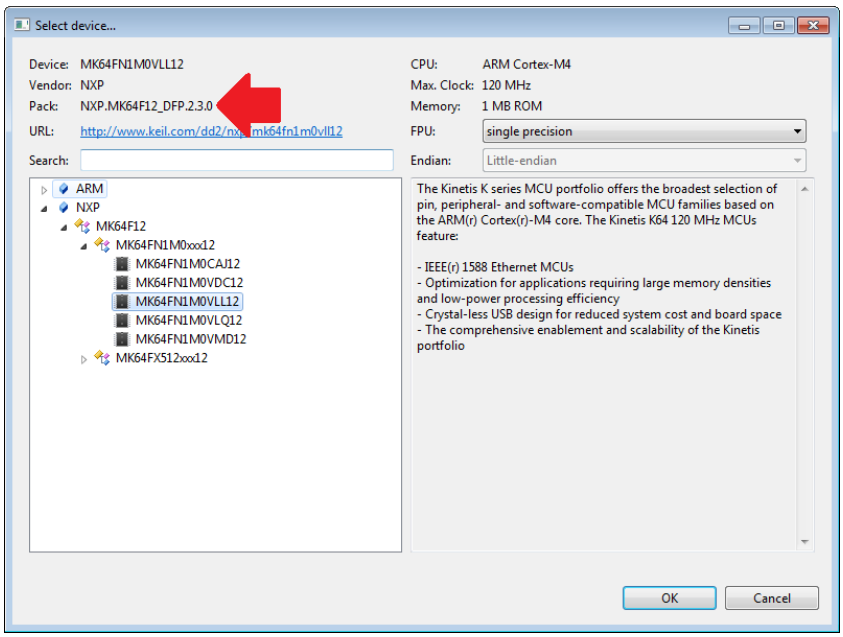



**Figure 40. Device from MCUXpresso SDK pack in IAR Embedded Workbench for Arm**

3. Go to *Project > CMSIS-Pack > Project Components* and unfold the Device class. Find *Startup* component, click on the *Vendor* column and switch the startup component from Keil to NXP. Click the *OK* button to close *Project CMSIS Component Manager*.

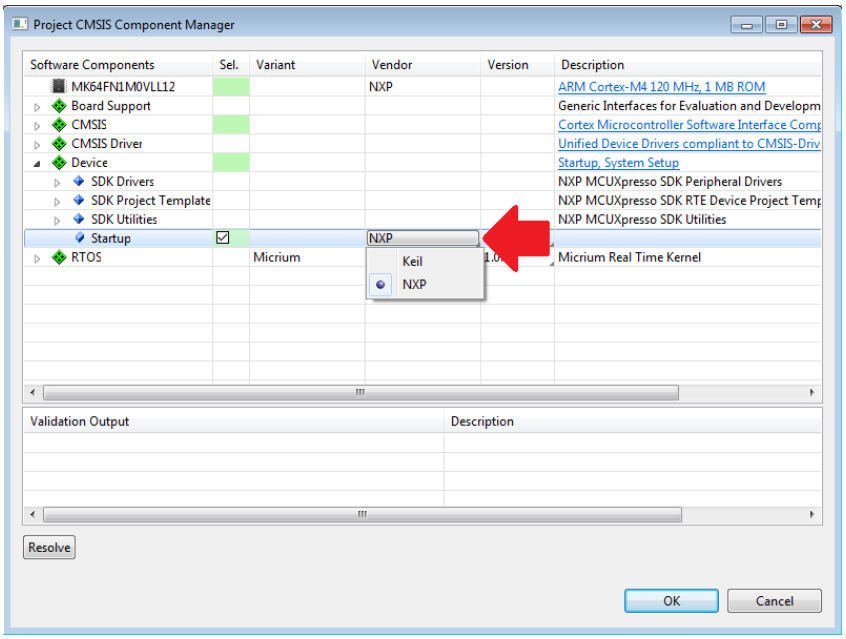



**Figure 41. Switching startup component in IAR Embedded Workbench for Arm**

4. In the Workspace Window you can see that Startup component files from NXP MCUXpresso pack replaced the files from Keil pack. Note that the files from Keil pack are still available in the *{project}\RTE\Kinetis_K60_DFP\Keil* directory.

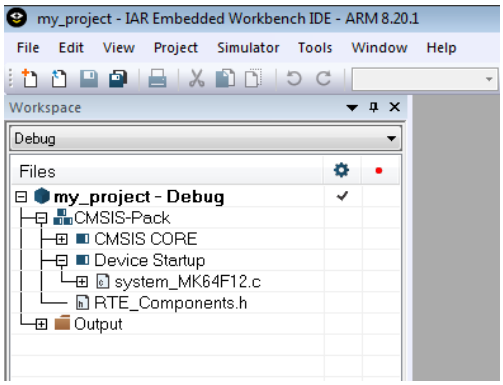



**Figure 42. Startup component from Keil pack in IAR Embedded Workbench for Arm**

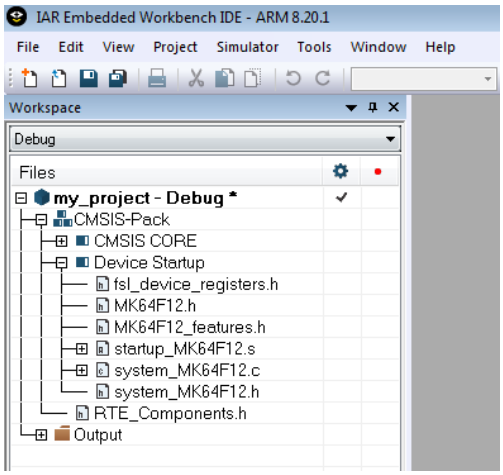



**Figure 43. Startup component from MCUXpresso SDK pack in IAR Embedded Workbench for Arm**

5. You can add more MCUXpresso SDK components using *Project CMSIS Component Manager*.

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